



Reviewing the issue of pollinator decline through an integrated emergency management

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ABSTRACT

Globally, habitat loss, pesticide use, climate change, and the spread of invasive species have all contributed to a significant decline in pollinator populations. This study hypothesizes that pollinator decline constitutes a societal emergency. This study reviewed strategies for improving pollinator decline through effective emergency management. The PRISMA 2020 checklist and flow diagram were utilized to conduct a qualitative descriptive analysis. Qualitative texts on pollinator decline, various perspectives, and emergency management were examined as part of the research methodology. This study identifies and categorizes multiple individualized approaches from international organizations, developed nations, and developing nations. All major stakeholders in the global arena can be classified into these three groups. The key finding is that these three stakeholders must adapt their individualized approaches through a framework of integrated emergency management to effectively mitigate pollinator decline. When formulating a global strategy, these stakeholders should prioritize global stewardship, alternative pollinators, pollinator-friendly practices, emergency training and exercises, and other related initiatives. Compared with earlier studies, this study provides a more comprehensive examination of how integrated emergency management principles can be applied to confront pollinator decline by including global stakeholders, strategies, geographical locations, and multi-disciplines.

1. Introduction

1.1. Basic information on pollinator decline

The pollinators discussed in this study include various insects (e.g., bees, butterflies, beetles, and moths) as well as some animals (e.g., birds, bats, reptiles, and mammals). They are responsible for pollinating approximately 60–70 % of all plants on Earth and facilitating the transfer of male pollen between flowers [1]. Other pollination methods include wind, water, and self-pollination. In particular, rice, maize, and wheat are primarily pollinated by wind, while many nuts and fruits (approximately 35 % of agricultural crops) rely on various animals and insects for pollination [2,3]. In addition, domesticated and wild bees (25,000 species) are recognized as particularly effective pollinators for agricultural crops.

Approximately 50 % of pollinators worldwide (approximately 200,000 species) are threatened by pollinator decline [4]. This decline reduces pollination across national boundaries and is caused by several interconnected factors, the most commonly cited being climate change, habitat loss, the spread of invasive species, outbreaks of animal diseases

and parasites, and intensive pesticide use. Collectively, these factors account for approximately 45 % of the recorded declines in pollinator populations [5–7]. In particular, human activities, such as the expansion of cultivated areas, changes in irrigation practices, and overgrazing, have steadily contributed to pollinator decline [8]. Addressing pollinator decline is more complex than previously understood [9].

Pollinator decline directly results in pollination deficits, which in turn lead to various adverse effects on human society. For instance, approximately half a million people face premature death annually due to the direct or indirect health impacts of pollinator decline [10]. The economic value of pollinators was estimated at USD 173 billion in 2009 and USD 577 billion in 2016 [11,12]. The economic loss attributed to pollinator decline in 2016 was estimated at USD 1600,191 billion [13]. The consequences of pollinator decline include ecosystem disruption, environmental impacts, and other less visible phenomena [14].

1.2. Previous literature

German researchers, including Joseph Koelreuter in the 18th century, began investigating the roles of insects in plant reproduction [15].

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Charles Darwin supported the theory that managed pollinators enhance fruit and seed production [16]. In 1998, numerous researchers discussed the pollinator decline at the First International Conference on the Decline of Pollinators in the Neotropics held in São Paulo, Brazil [17]. While few researchers regard pollinator decline as an ecological calamity, several fields—such as pollination biology, floral biology, entomology, and environmental studies—have advocated for urgent alternatives [18,19].

Two independent studies reported contrasting findings regarding the nutritional impact of animal pollinators on humans. According to Eilers et al. (2011), crops that relied entirely or partially on animal pollinators contributed significantly to human nutrition by giving vital elements including lipids, lycopene, folic acid, vitamins A and C, and others [20]. Consequently, they advocate the role of animal pollinators in human nutrition. By contrast, Ellis et al. (2015) reported that while vitamin A could enhance human nutrition, the overall role of animal pollinators depended on local dietary habits [21]. In short, these studies do not fully support the significance of animal pollinators.

Three separate studies examined the effects of neonicotinoid insecticides and proposed modern management solutions. Goulson's research emphasized that pollinators are among the many non-target species adversely affected by neonicotinoids [22]. Rundlof et al. indicated that neonicotinoids posed an intolerable risk to bees [23]. Pott et al. emphasized that effective management strategies could safeguard both managed and wild pollinators [24].

The assessment study on pollinator decline released by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) is considered a seminal work in the field [25]. It underscored the economic and social significance of pollinators, along with the increasing risks from human activities. To mitigate further pollinator loss, the report examined various management and policy approaches, stressing the importance of independent, evidence-based evaluations to address these complex issues fairly.

Bartomeus and Dicks (2019) investigated the importance of a coordinated research infrastructure for pollinators [26]. They posited that the lack of accessible pollinator data was a significant challenge and advocated for the integration of ecological, economic, and social research databases, along with interdisciplinary research. Although their focus was on research infrastructure, they did not fully address the comprehensive aspects of pollinator decline.

The Federal Emergency Management Agency (FEMA) published guidelines on nature-based solutions [27]. These approaches employ natural features and processes to address various societal challenges, including biodiversity loss. Specifically, FEMA recommended incorporating natural elements (e.g., trees, animals, and fish) into traditional infrastructure to mitigate the effects of natural disasters, including climate change. However, pollinator decline in local communities was not addressed in the FEMA advisory.

1.3. Aim of the study

Pollinator decline is a widespread and diverse issue globally [28,29]. North American countries face a high risk of pollinator decline, while European countries experience moderate instability in crop yield. By contrast, many countries in South America, Africa, Asia-Pacific, and Australia regard the situation as particularly severe. In general, nations in the Global South perceive the core issue of pollinator decline as more severe than those in the Global North [30,31].

Humans continue to depend on the cultivation of plants, which in turn rely on pollinators. Therefore, maintaining a balance is crucial for the survival of the human race [32,33]. While pollinator decline would not cause the complete disappearance of plants worldwide—owing to wind, water, and self-pollination—it has disrupted not only plants, animals, and insects but also human society, with profound implications.

The occurrence of pollinator decline in the 21st century has triggered multiple societal emergencies, given that 35 % of leading food crops

depend on pollinators [34–37]. Consequently, international organizations and nations must systematically enhance their strategies to address pollinator decline, particularly through comprehensive approaches. This study seeks to answer the research question: “How have major stakeholders addressed the issue of pollinator decline through emergency management?” The stakeholders considered are international organizations, developed nations, and developing nations.

The purpose of this research is to describe the current status of pollinator decline and provide recommendations to reduce human mortality and economic impacts. Various individualized approaches and integrated emergency management strategies for pollinator decline employed by international organizations, developed nations, and developing nations are analyzed. Specifically, the Results section examines the status of multiple individualized approaches worldwide, while the Discussion section will offer corresponding recommendations.

1.4. Characteristics of the study

Multiple researchers have investigated the biological and entomological aspects of pollinator decline; however, its impact on human society remains inadequately explored [38]. Moreover, no study has yet used the integrated emergency management concept to link ecological, social, and policy aspects within a single framework in order to address the complex issues that pollinator decline presents, including disruptions to food security, ecosystem stability, and agricultural productivity. This research aims to comprehensively describe how human society can respond to pollinator decline through integrated emergency management.

This study proposes two distinct approaches: multiple individualized approaches and integrated emergency management. The term “multiple individualized approaches” refers to the fragmented efforts addressing pollinator decline that remain uncoordinated and often scattered across various regions [39]. Here, “individualized” highlights how these efforts are compartmentalized, separated, self-directed, and limited worldwide. This approach encompasses the negative features associated with the current baseline of pollinator decline.

In contrast, integrated emergency management fosters collaboration among all stakeholders in a region during emergency situations [40]. Applying this strategy to pollinator decline signifies a unified effort among stakeholders through networking. Here, the term “integrated” emphasizes mutual, interconnected, and contextually interdependent management in response to pollinator decline emergencies. Unlike multiple individualized approaches, this strategy offers a positive and coordinated impact on addressing pollinator decline.

2. Methods

A central methodology in this study is qualitative descriptive analysis, which identifies and interprets qualitative text data. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist and flow diagram have facilitated the qualitative descriptive analysis (refer to the Data availability statement). This study aims to identify the major stakeholders and their actions regarding pollinator decline, necessitating qualitative descriptive analysis. The methodology comprises four steps: establishing the research design, collecting qualitative data, analyzing qualitative data, and presenting qualitative data [41]. Fig. 1 illustrates the research design framework, summarizing the research question, coding (e.g., allocating numbers to excerpts from original qualitative texts), units of descriptive analysis (i.e., three stakeholders), descriptive-analytical categories (i.e., two major approaches), and other relevant functions.

For qualitative data collection, this research utilized major search engines, including Google, ScienceDirect, EBSCO, ProQuest, the Military Database, and Oxford University Press. Search terms included “pollinators,” “pollinator decline,” “pollinators and emergency management,” and “pollinator decline and human society,” among others. In addition,

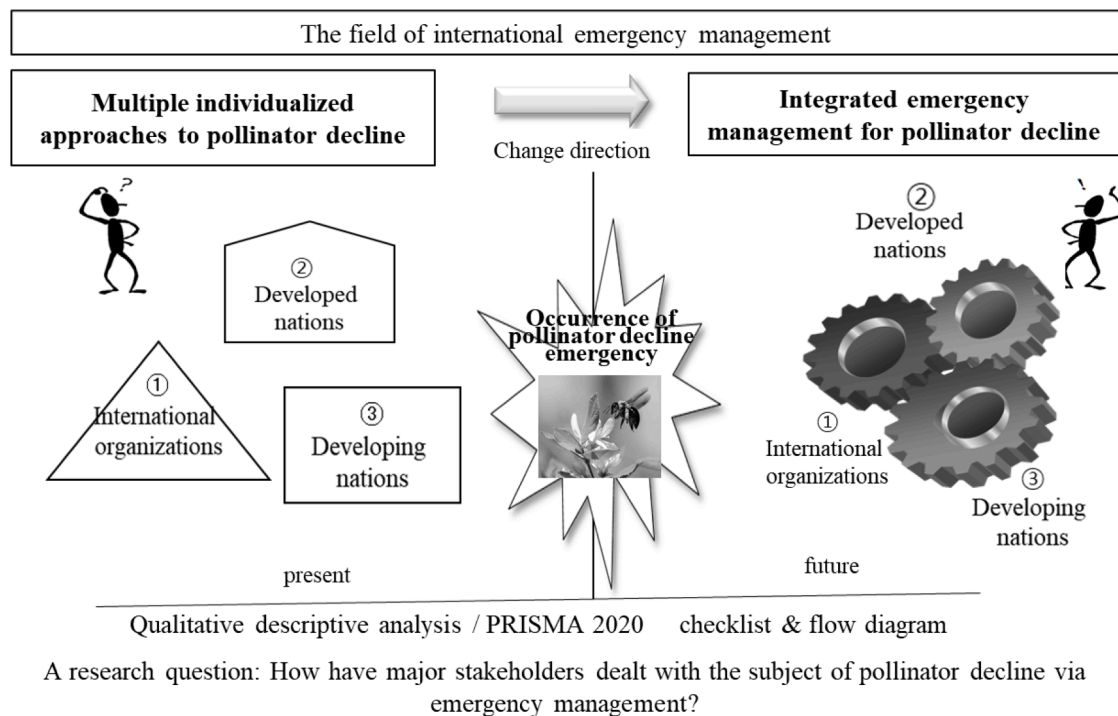


Fig. 1. Research design framework.

this study partially relied on triangulation to validate the identified qualitative texts. Examples include using a number of databases to increase the reliability of data collection (data triangulation) and interpreting results using a variety of theoretical frameworks, such as integrated emergency management, ecological theory, and environmental theory (theory triangulation) [42]. Also, the PRISMA 2020 checklist and flow diagram's organized application offers enough rigor, openness, and reproducibility to make up for this drawback.

Google provided access to various government documents and research articles on pollinator decline, while ScienceDirect offered numerous scientific articles on the topic. EBSCO and ProQuest contributed a limited number of studies on the relationship between pollinator decline and human activity. Inclusion and exclusion criteria were based on the presence of information regarding pollinator decline, individualized approaches, integrated emergency management, or the human perspective. Accordingly, a result, 80 materials in all—N including 5 books, 9 official documents, 51 research articles, and 15 more online sources—were used in this study.

In summary, although this study did not formally consult a librarian while developing the search strategy, it adhered to established practices in qualitative descriptive analysis, which is synonymous with systematic literature review [43]. Core concepts guided the iterative construction of search terms. The selected databases were chosen for their coverage of various multidisciplinary topics, including ecology, environmental science, and emergency management. This approach ensured a focus on literature addressing the ecological aspects of pollinator loss, as well as its management, policy implications, and societal consequences.

For qualitative data analysis, three units of descriptive analysis were selected and used during coding. These included all important international emergency management stakeholders involved in pollinator decline. For example, international organizations such as the United Nations (UN) have provided guidelines on this issue for multiple nations. Countries were also classified into two groups based on their economic status: developed and developing. In summary, these three stakeholders encompass the key players in addressing pollinator decline.

To elaborate, Relevant text passages were given three codes that reflected their main themes in order to provide further detail. For

traceability, each coded snippet was assigned a number (e.g., S1, S2, and S3), which allowed for a clear connection between the analysis and the original content. After grouping excerpts with the same coding, patterns were found and then combined to create descriptive themes [44]. This method preserved the data's original meaning while enabling a clear and methodical analysis of the contents.

Various factors—including the literature review, search strategy, and other relevant considerations—were considered when determining the specific contents of each unit of descriptive analysis. Among these, the literature review had the most significant influence on the content [45]. While the study aimed to incorporate qualitative texts from all six continents, it ultimately focused on a limited number of case studies. The necessity to maintain balance among the three descriptive analysis units was the primary motivation behind this decision [46].

In presenting qualitative data, the study aimed to record appropriate interpretations of pollinator decline across designated sections [47]. Considering the close relationship between the two analytical categories and the study's results, the analysis concentrated on describing how each analytical category appeared in the field. The analytical category of multiple individualized approaches to pollinator decline encompasses uneven efforts, while integrated emergency management promotes for balanced suggestions.

3. Results: multiple individualized approaches to pollinator decline

Subjective interpretation during study selection, topic classification, and data extraction has often introduced bias in PRISMA-based qualitative descriptive analyses [48,49]. These arbitrary decisions can affect the prioritization of studies, synthesis of data, and conclusions drawn. However, the employment of the PRISMA 2020 checklist and flow diagram has helped mitigate these issues. This framework has encouraged clear eligibility criteria, systematic and repeatable search methods, and transparent documentation of study inclusion and exclusion. In this regard, Table 1 offers comprehensive details on the methods for data collection and analysis covered in this part.

Table 1

Text data collection and analysis in the section of results.

Units	Text types	No. of included texts	Coded excerpt	Resulting theme
3.1 International organizations	- Official documents	- 1	“Favourite pollinator protection has been given top priority by many international organizations in the larger context of biodiversity conservation.”	Multiple individualized approach
	- Research articles	- 2		
	- Online sources	- Total: 4		
3.2 Developed nations	- Official documents	- 1	“Transboundary pollination issues have received little attention from many high-income countries, which have concentrated on their domestic contexts.”	Multiple individualized approach
	- Research articles	- 1		
	- Online sources	- Total: 3		
3.3 Developing nations	- Books	- 1	“With little participation in cooperative research projects, many low-income countries have relied on traditional pollination methods.”	Multiple individualized approach
	- Research articles	- 3		
		- Total: 4		

3.1. International organizations

Few UN agencies have addressed pollinator decline, including the UN Environment Programme (UNEP), the Food and Agriculture Organization (FAO), the UN Educational, Scientific and Cultural Organization (UNESCO), and the UN Development Programme (UNDP) [50]. Each agency focused on specific aspects of pollination, including environmental pollution, food security, local knowledge, and conservation. However, UN efforts to combat pollinator decline have yet to evolve into coordinated actions among its agencies [51].

Although several international non-governmental organizations (INGOs) have engaged with pollinator decline, many have limited their activities to specific pollinators [52]. Examples include the Bee Parks Trust (bees in southern Africa), the Bombay Natural History Society (moths in India), Grassroots Alliance PERESVET (bats in Russia), and BEES for the World (bees in Germany) [53]. Although such specialized areas facilitate focused initiatives, this approach leaves other pollinators underrepresented, thereby widening the gap among them.

3.2. Developed nations

The European Union (EU) has initiated various programs and national strategies on pollinator decline through agriculture, pesticide regulation, environmental protection, and scientific research. However, most of these initiatives have not been effectively monitored [54]. Notably, EU programs have lacked consistency and have failed to incorporate effective tools and operational mechanisms to address pollinator decline [55]. Significant gaps exist among policymakers, program leaders, and other stakeholders, reflecting a lack of actionable criteria.

Other developed countries have adopted individualized approaches to pollinator decline. For instance, the United Arab Emirates (UAE) has relied heavily on honeybees in its harsh desert environment and thus recently introduced the Emirati Queen Bee for food production [56]. The country has even employed drones for pollination, although these have not been effectively utilized for other pollinators. Australia faces challenges in distinguishing between pollinator decline, species migration, and normal species movement, resulting in limited information on pollinators, including moths.

3.3. Developing nations

While defying food insecurity, an increasing number of African countries, such as Ghana, Kenya, Burundi, and Zambia, have joined international networks to combat pollinator decline, including the African Pollinator Initiative and the Coalition of the Willing on Pollinators [57]. In addition, some nations, such as Nigeria, have developed national strategies to address pollinator decline. However, collaborative research among multiple countries remains fragmented, and many

African programs lack cohesion [58].

Other developing nations, such as Brazil in South America and India in South Asia, exhibit their own patterns of pollinator decline. Despite Brazil's status as the most biologically diverse nation globally, 88 % of its bees and 95 % of its vertebrate pollinators have been overlooked [59]. Similarly, while certain local fruits in India, including mango, guava, banana, and cocoa, are commonly pollinated by bats, there is insufficient information and data regarding these nectar-feeding bats [60].

4. Discussion

Local residents and institutions often struggle to grasp the exact situation of pollinator decline within their communities, as even well-studied areas possess limited information on the subject [61,62]. Many individuals are unaware of which pollinators are declining, their distribution, the number of alternative pollinators available, or what alternatives should be provided. Consequently, the need for integrated approaches to address pollinator decline has become increasingly urgent worldwide.

Although the impacts of pollinator decline have been partially recognized in various regions, the extent of associated risks has varied significantly [63,64]. Local residents and institutions perceive the critical risks stemming from pollinator decline differently. This risk perception is not objective; rather, it is subjective and influenced by the rational knowledge and emotional appraisals of individuals and institutions.

Developing initiatives to combat pollinator decline may initially appear simpler when stakeholders adopt individualized approaches, as these methods can sidestep complex challenges. However, such tailored strategies may face difficulties when the number of pollination programs increases, leading to duplication of efforts, redundancy in research findings, fragmented policy guidelines, lack of accountability, potential exhaustion, and ease of implementation [65]. These factors indicate that relying solely on individualized strategies may ultimately diminish overall effectiveness.

In reality, pollinator decline is not completely attributable to the occurrence of a natural hazard or a manmade crisis; it is a societal emergency [66] that affects all facets of human society. Therefore, addressing the issue of pollinator decline must be integrated across multiple domains, including human resources, public health, agricultural development, food security policy, natural resource management, biodiversity conservation, the chemical industry, and the electronics sector.

Integrated emergency management considers the interplay between humans, plants, insects, animals, and other environmental factors [67]. Consequently, the decline of pollinators cannot be considered in isolation from broader ecological issues. Instead, pollinator health is enhanced through reciprocal relationships not only with humans but

also with a diverse array of insects, trees, animals, and environmental elements.

From a human perspective, integrated emergency management aligns with global stewardship efforts to address pollinator decline [68]. Global stewardship, as an environmental approach, emphasizes long-term sustainable growth. It invites participation from various stakeholders, including accredited scientists, policymakers, journalists, environmentalists, and local residents. This collaborative approach encompasses both intrinsic challenges and solutions, as well as extrinsic values and philosophies.

The quality of pollinator habitats and the complementarity and connectivity among individuals, nations, and nature must be improved to solve the problem of pollinator decline. Stakeholders should not rely on a single solution but should flexibly employ various strategies [69]. Moreover, as no single stakeholder can address the imminent threat of pollinator decline alone, all three stakeholder groups must shift from individualized approaches to integrated emergency management, as illustrated in Table 2.

The recommendations above are based on research and insights from certain chosen case studies that illustrate how different regions are responding to pollinator decline. To reflect variations in strategies, resources, and implementation capabilities, the three stakeholder groups have been included [72,73]. The case studies were chosen for their valuable examples of challenges and lessons learned related to these recommendations. Each suggestion addresses identified issues, ensuring that the proposed integrated emergency management approach is both comprehensive and practical.

The transition to integrated emergency management offers significant advantages for addressing pollinator decline. Primarily, it will help the three stakeholder groups to develop a comprehensive strategy for pollinator decline [74]. Integrated efforts will deepen stakeholders’

understanding of the complex nature of pollinator decline. Furthermore, fostering networks among stakeholders will enhance collaboration between national and international initiatives, ultimately leading to improvements in pollinator health.

This transition will yield additional positive effects for pollinators. For instance, integrated emergency management will enhance disaster resilience against pollinator decline in various regions by fostering connections among stakeholders [75]. Communities that are well-connected are significantly more adept at recovering from pollinator decline compared to affluent communities. Furthermore, the collaboration among the three stakeholders will boost workplace productivity by facilitating quick decision-making, streamlining processes, improving cost efficiency, reducing conflicts, and enhancing workplace culture.

The existing literature on pollinator decline and emergency management has frequently emphasized the need for comprehensive research on appropriate alternatives [76]. The introduction of integrated emergency management in addressing pollinator decline will bridge gaps in the current literature, thereby making a substantial contribution. By systematically applying the principles of integrated emergency management, the discourse on pollination will broaden its theoretical framework regarding pollinator decline. Likewise, the literature on emergency management will expand its focus to include detailed factors related to pollination.

Based on contributions to the literature, the transition will continue to advocate for the role of alternative pollinators [77]. For example, when honeybee populations decline, as observed in many countries, stakeholders will strive to recover their numbers. However, this does not signify the end of alternative pollination alternatives. Major stakeholders will persist in increasing the number of bees utilized for pollination. Indeed, they must prove whether these managed pollinators may pose risks [78].

Similarly, the transition will implement pollinator-friendly practices [79]. This category of pollinator-friendly practices comprises many sub-strategies, including planting native flora, creating habitats for pollinators, establishing butterfly highways, turning off outdoor lighting, reducing the use of lawnmowers, exhibiting pollinators, monitoring their migration, and enhancing public relations. Concurrently, all stakeholders must seek to identify new instances of pollinator-friendly practices in an ever-evolving environment.

The emerging topic of pollinator decline emergencies should be included in the scope of emergency training and exercises, such as the Exercise Bee Prepared initiative in Australia [80]. Emergency trainers can provide not only training on pollinators but also tools for identifying invasive species. Multiple trainees will learn to collaborate effectively while navigating partnerships around pollinators. Various types of emergency training and exercises, including seminars, interactive games, and field drills, may be utilized.

5. Conclusion

A widespread decline in pollinators will render our planet less vibrant and more desolate. Therefore, it is crucial to make regions with the presence of pollinators. This study aimed to explore strategies for improving the status of pollinator decline through the collaborative efforts of international organizations, developed nations, and developing nations. By presenting two significant models to include multiple individualized approaches and integrated emergency management, this study has achieved its primary goal. This will undoubtedly contribute to reducing human deaths, economic losses, and the psychological impacts associated with ongoing pollinator decline.

The most significant finding of this study is that all three stakeholders must work collaboratively to transform multiple individualized approaches into integrated emergency management strategies for addressing pollinator decline. For effective implementation, all three stakeholders should sequentially undertake the aforementioned tasks,

Table 2
Application of integrated emergency management to pollinator decline.

Analytical units	Specific alternatives
4.1. International organizations	- The UN Office for Disaster Risk Reduction (UNDRR), a leading UN emergency management agency, will coordinate the individualized strategies of UN agencies addressing pollinator decline. Aligning these initiatives with the Sustainable Development Goals (SDGs), particularly SDG 12 (Responsible Consumption and Production) and SDG 15 (Life on Land), will help guarantee that collective efforts not only tackle current threats but also promote ecological sustainability and resilience in the long term [70]. - Numerous international non-governmental organizations (INGOs) must forge extensive partnerships with one another and other organizations to facilitate information sharing and strategy development targeting specific pollinators.
4.2. Developed nations	- The European Union (EU) must integrate robust tools and control mechanisms to address the breadth of pollinator decline by systematically assigning roles and responsibilities to relevant stakeholders. - As the UAE expands its pollination efforts to include local pollinators such as Arabian Purple Sunbirds and honeybees, Australia will address pollinator decline through an enhanced classification of natural resources.
4.3. Developing nations	- Many African nations should design and implement coordinated joint research and regional programs focused on pollinator decline while establishing mutual adjustment practices [71]. - Both Brazil and India will systematically gather and redistribute scattered information on neglected pollinators to local communities to enhance public perception of these vital species.

Notes: The chosen cases in Table 2 were considered as the most typical instances of integrated emergency management to pollinator decline in the studied literature. These instances were selected because of their worldwide implications and data availability.

such as coordinating efforts through UNDRR, assigning roles and responsibilities within the EU, and defining mutual adjustment practices among African nations during joint research. Moreover, these stakeholders will benefit from the positive outcomes of integrated emergency management, a comprehensive approach, potential alternative pollinators, pollinator-friendly practices, and training exercises.

This study has particularly emphasized how fragmentation and inadequate coordination can hinder the efficacy of multiple individualized approaches. The examination of these various strategies reveals both exemplary practices—such as targeted local interventions informed by ecological data—and less effective practices, such as isolated efforts that lack scalability or coordination with larger initiatives. These revelations highlight how using an integrated emergency management framework to provide a more robust and cohesive strategy for addressing complex ecological challenges.

This study has comprehensively focused on integrated emergency management in relation to pollinator decline more comprehensively than existing research. Consequently, it has explored the relationship between pollinator decline and human society. However, a limitation of this study is the difficulty in accurately assessing the situation of pollinator decline in certain regions, given the lack of data on the distribution of endangered species. In addition, local residents' perceptions of risk regarding pollinator decline have not been objective but rather subjective (e.g., geographical bias).

Accredited researchers will further apply the principles of emergency management to the issue of pollinator decline, including integrated emergency management, comprehensive emergency management, and concepts of progressiveness and professionalism, by referencing the framework established in this research. By addressing the limitations identified in this study, researchers will seek to mitigate subjective risk perceptions and geographical biases surrounding pollinator decline. In many senses, a united pollinator emergency framework in line with international accords is desperately needed to handle the growing catastrophe, promote coordinated international cooperation, and ensure direct policy relevance. These efforts will continue to bridge the knowledge gaps between pollinator decline and human society.

CRediT authorship contribution statement

Kyoo-Man Ha: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The author declares that he has no financial or non-financial conflict of interests with this manuscript.

Data availability

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